Sepand AliMadadSoltani

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Education

Master 2 in Health Engineering: Medical Device Engineering Lyon, France
Polytech Lyon, Claude Bernard University Lyon 1 2024-2025

- **Concentration:** Medical Image Processing
- **GPA:** 15.31/20
- **Courses:** Magnetic Resonance Imaging (MRI), Segmentation & Registration, Artificial Intelligence in Medical Imaging, Image Reconstruction & Inverse Problems
- **€10,000** Excellence Scholarship awarded for excellent academic background

Bachelor of Science in Electrical Engineering Tehran, Iran
K.N. Toosi University of Technology 2018-2023

- **Concentration:** Biomedical Engineering
- **GPA:** 16.33/20
- **Courses:** Statistical Pattern Recognition, Signals & Systems, C Programming, Engineering Math, Engineering Probability

Research Experience

Master's Internship: Bayesian Inference of Image-Derived Input Function in Dynamic PET/MR Imaging Lyon, France
CERMEP Laboratory March 2025-August 2025

- Developed non-invasive alternative to arterial sampling for PET quantification in 59 patients
- Built an accurate and automatic carotid artery segmentation pipeline on TOF-MR angiography images
- Implemented Bayesian inference model using Markov Chain Monte Carlo (MCMC) for Partial Volume Correction
- Reduced quantification bias significantly compared to standard image-derived methods (MAPE 13% vs 24%)
- Validated method using Monte Carlo PET simulations and cross-site collaboration with Paris-Saclay

Bachelor's Thesis: Development of a Interactive and Intelligent Tissue Segmentation Application Tehran, Iran
Machine Vision & Medical Image Processing Laboratory (MVMIP), KNTU January-June 2023

- Built medical image analysis and visualization software in Python
- Implemented intelligent scissors algorithm for semi-automatic boundary detection on 2D image slices
- Integrated the algorithm into the app for fast tissue segmentation with minimal user input
- Incorporated complementary segmentation tools (polygon selection, thresholding, otsu filter)

Work Experience

Sharif University Science & Technology Park Tehran, Iran
C++ Software Developer October 2023-July 2024

- Developed custom QML components to render massive high-frequency datasets at 60 FPS without UI blocking.
- Designed a multi-threaded data acquisition backend to interface with PCIe hardware drivers.
- Implemented real-time digital signal processing (DSP) algorithms using Eigen and Boost
- Embedded a Python interpreter via pybind11 to execute ML inference on C++ data streams.

TECVICO Vancouver, Canada (Remote)
Medical Imaging Python Developer July-September 2023

- Engineered a medical image visualization tool (PyQt, VTK, ITK) with thresholding, segmentation, and annotation capabilities.
- Created a dynamic node-based Qt UI to construct and execute image processing pipelines.
- Integrated AI algorithms into the processing pipeline in collaboration with the R\&D team.

Skills

- **Programming:** C++, Python, MATLAB, Bash
- **Software and Tools:** GNU/Linux, Git, FMRIB FSL, SPM, 3D Slicer, TPCCLIB
- **Libraries:** Tensorflow, PyTorch, NumPy, pandas, scikit-learn, Matplotlib, ITK, VTK, Boost, Qt
- **Languages:** English (C2), French (B2), Persian (Native)
- **Standards** IEC 62304, ISO 14971, ISO 13485

Conference Posters

Bayesian Image-Derived Input Function Estimation for [^{18}F]DPA-714 PET: An Evaluation Study
Submitted Abstract (Under Review)
Chloé Teurquety, **Sepand Alimadadsoltani**, et al.
Submitted to The Neuroreceptor Mapping 2026 Conference , June 2026

Improved Noninvasive Quantification of PET Kinetics Using Bayesian Geometric Transfer Matrix (BGTM)
Submitted Abstract (Under Review)
Inés Mérida, **Sepand Alimadadsoltani**, et al.
Submitted to The Neuroreceptor Mapping 2026 Conference , June 2026

Projects

Image-based Persian and English Character Sequence Recognition using Recurrent Convolutional Neural Networks(RCNN)

- Implemented the network based on a paper using the Tensorflow library in Python
- Synthesized images of Persian text of different variety
- Applied data augmentation techniques such as rotating, translating, adding distortion, and adding noise to images
- Successfully trained the model for both languages using the self-made synthesized Persian dataset and public English datasets
- Achieved +85% accuracy for both languages

Deep Learning Analysis of Functional Connectivity Maps in Alzheimer's Disease

- Processed raw fMRI/MRI data from the ADNI database using FSL (motion correction, registration) to extract functional connectivity maps
- Evaluated RCNN and CNN models in Tensorflow for feature extraction to classify Alzheimer’s disease stages
- Identified critical limitations in using standard deep learning architectures for fMRI analysis

Automated fMRI Pre-processing Pipeline

- Implemented brain extraction from structural reference MR image
- Implemented fMRI pre-processing including motion correction, slice timing correction, spatial smoothing, and co-registration
- Optimized the pipeline with parallel processing to accelerate computation on the ADNI dataset

The Game of Tetris with a Custom Game Engine Using OpenGL in C++

- Developed a custom 2D graphics renderer completely from scratch using the OpenGL graphics API in C++
- Implemented user input handling, navigatable menus, and text rendering capabilities to the engine
- Designed and implemented the game of Tetris using the said engine in Object Oriented C++

Rendering the Shadow of a 4D Hypercube in 3D using Qt/VTK/C++

- Programmed the geometric projection of a 4D hypercube onto 3D space to visualize higher-dimensional structures
- Achieved real-time rendering and interactive manipulation by wrapping VTK within a Qt GUI

Modeling the Magnetic Field of a Spherical Solenoid in MATLAB

- Derived the analytical magnetic field equations for a spherical solenoid
- Implemented numerical computations in MATLAB to calculate the field across multiple planes
- Designed an interactive GUI using MATLAB App Designer to visualize 3D heatmaps and vector plots